

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Design a Basic Webpage Using HTML5 and Internal CSS3 (Header and Content Containers)

2. AIM

To design and develop a basic webpage using HTML and CSS with a styled header container and a styled content container, using internal CSS written inside the `<style>` tag.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML webpage.
- To use the `<div>` element to group content into containers.
- To apply internal CSS using the `<style>` tag inside the `<head>` section.
- To style the page body using font-family, background-color and padding properties.
- To style containers using background colors, text colors and padding.
- To use class selectors (`.header-box` and `.content-box`) to style specific elements.
- To use the margin-top property to create spacing between two containers.
- To align text using the text-align property.
- To understand the separation of content (HTML) and presentation (CSS).

4. THEORY

HTML (HyperText Markup Language) is used to define the structure and content of a webpage. Elements such as `<html>`, `<head>`, `<body>`, `<div>`, headings (`<h1>`, `<h2>`) and paragraphs (`<p>`) are used to organize the content.

CSS (Cascading Style Sheets) is used to control the presentation and appearance of HTML elements such as colors, fonts, spacing and alignment. CSS can be applied in three ways: inline, internal and external. In this experiment, **internal CSS** is used, where the styles are written inside a `<style>` tag in the `<head>` section of the same HTML file.

The webpage titled "**Very Basic Page**" consists of:

- **Body styling:** Arial font, light gray background (`#f0f0f0`), 20px padding, and display: inline-block.
- **Header container (.header-box):** Blue background, white text, 15px padding, centered text; contains the heading "My Website" and a paragraph.

- **Content container (.content-box):** White background, 20px padding, 15px top margin; contains the heading "About This Page" and a paragraph.

Important CSS concepts used are the **class selector** (.classname), which styles all elements carrying that class; the **box model** (padding and margin), where padding adds space inside a box and margin adds space outside it; and **display: inline-block** on the body, which makes the body shrink to fit its content width. The property flex-direction: column is also written on the body, but it has no effect because the body is not a flex container (display: flex is not applied).

The basic working flow is:

HTML Structure → Internal CSS Styling → Browser Rendering → Styled Webpage

5. ALGORITHM / PROCEDURE

1. Create a new project folder for the webpage.
2. Create an HTML file named index.html.
3. Write the <!DOCTYPE html> declaration and the <html> root element.
4. Inside <head>, add the page title "Very Basic Page".
5. Add a <style> tag inside <head> to write the internal CSS.
6. Style the body with Arial font, background color #f0f0f0, 20px padding and display: inline-block.
7. Create the .header-box class with blue background, white text, 15px padding and centered text.
8. Create the .content-box class with white background, 20px padding and 15px top margin.
9. Inside <body>, create a <div> with class "header-box" containing an <h1> heading and a <p> paragraph.
10. Create a second <div> with class "content-box" containing an <h2> heading and a <p> paragraph.
11. Add HTML comments to mark the Header Div and Content Div.
12. Save the file as index.html.
13. Open index.html in a web browser and verify the output.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>

<head>
  <title>Very Basic Page</title>
  <style>
    /* Basic body styling */
    body {
      font-family: Arial, sans-serif;
      background-color: #f0f0f0;
      padding: 20px;
      display:inline-block;
      flex-direction: column;

    }

    /* Styling for the header container */
    .header-box {
      background-color: blue;
      color: white;
      padding: 15px;
      text-align: center;
    }

    /* Styling for the content container */
    .content-box {
      background-color: white;
      padding: 20px;
      margin-top: 15px;
    }
  </style>
</head>

<body>

  <!-- Header Div -->
  <div class="header-box">
    <h1>My Website</h1>
    <p>Lorem ipsum dolor sit amet consectetur adipisicing elit. Voluptatibus minima ullam unde, nostrum, molestias
      est enim, deleniti cumque aliquam laudantium temporibus! Facere nihil perferendis pariatur blanditiis.
      Soluta corrupti dolorem fugit?</p>
  </div>

  <!-- Content Div -->
  <div class="content-box">
    <h2>About This Page</h2>
    <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit. Voluptate in incidunt tempora nemo dicta dolorem
      alias? Libero praesentium, unde odit itaque natus eum omnis id, molestias nesciunt non error nostrum?</p>
  </div>

</body>

</html>
```

7. CODE EXPLANATION

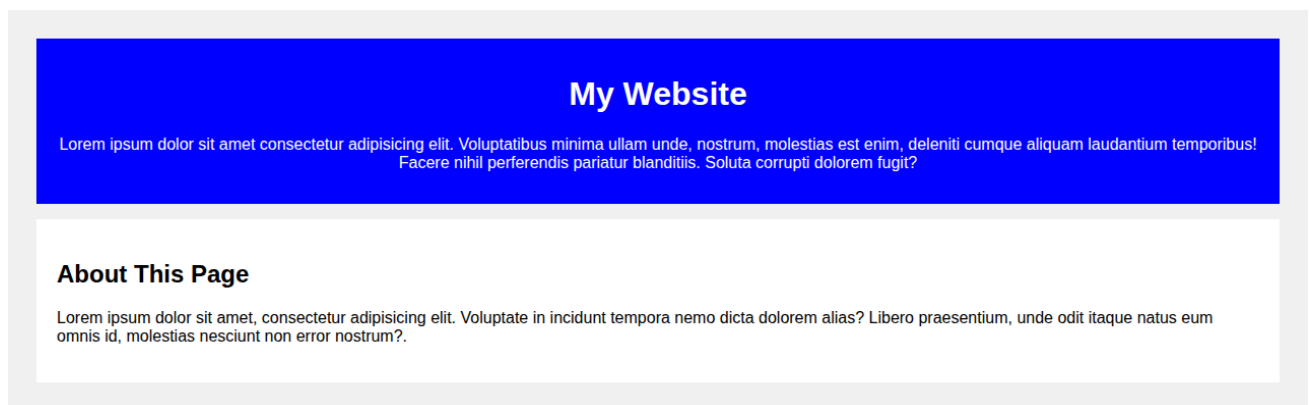
Code	Explanation
<!DOCTYPE html>	Declares the document as an HTML5 document.
<html>	Root element that wraps the entire webpage.
<head>	Holds page information such as the title and internal CSS.
<title>	Specifies the title "Very Basic Page" shown in the browser tab.
<style>	Contains the internal CSS used to style the page.
/* ... */	CSS comment; ignored by the browser.
body	Selector that styles the whole page body.
font-family	Sets the font to Arial, falling back to sans-serif.
background-color	Sets the background color of an element (#f0f0f0 for body, blue for header, white for content).
padding	Adds space inside an element between its content and border.
display: inline-block	Makes the body shrink to fit its content instead of stretching as a block.
flex-direction: column	Has no effect here because the body is not a flex container.
.header-box	Class selector for the header container.
color	Sets the text color (white inside the header).
text-align: center	Centers the text horizontally inside the header.
.content-box	Class selector for the content container.
margin-top	Adds 15px of space above the content box, separating it from the header.
<body>	Contains all visible content of the webpage.
<!-- ... -->	HTML comment used to label sections of the code.
<div class="...">	Groups related elements into a container and links it to a CSS class.
<h1>, <h2>	Heading elements ("My Website" and "About This Page").
<p>	Paragraph element that holds the text content.

8. TEST CASES AND EXECUTION DATA

The webpage was executed in a web browser and tested for page loading, header styling, content styling, body styling, spacing between containers, text alignment, and behavior on a narrow browser window.

Test Case	TC-01
Test / Input	Load the webpage in the browser
Expected Result	The page should display the blue header container and the white content container on a gray background.
Actual Result	Both containers were displayed correctly on the gray page background.
Status	Pass

TC-01 Output:



Test Case	TC-02
Test / Input	Check the header container
Expected Result	The header should have a blue background, white text, and show the heading "My Website" with its paragraph.
Actual Result	The header box was displayed with blue background, white text, and the heading and paragraph were shown correctly.
Status	Pass

TC-02 Output:



Test Case	TC-03
Test / Input	Check the content container
Expected Result	The content box should have a white background and show the heading "About This Page" with its paragraph.
Actual Result	The content box was displayed with white background and the heading and paragraph were shown correctly.
Status	Pass

TC-03 Output:

About This Page Lorem ipsum dolor sit amet, consectetur adipisicing elit. Voluptate in incididunt tempora nemo dicta dolorem alias? Libero praesentium, unde odit itaque natus eum omnis id, molestias nesciunt non error nostrum?.
--

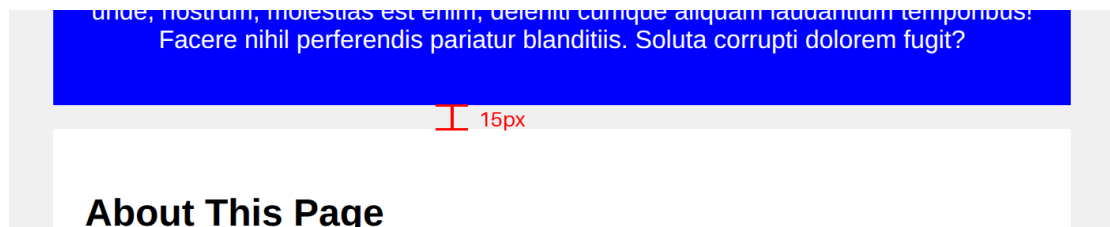
Test Case	TC-04
Test / Input	Check the body styling (font, background and padding)
Expected Result	The page should use Arial font, a light gray (#f0f0f0) background, and 20px padding around the containers.
Actual Result	Arial font and light gray background were applied, and the containers were spaced away from the page edge by the body padding.
Status	Pass

TC-04 Output:

My Website Lorem ipsum dolor sit amet consectetur adipisicing elit. Voluptatibus minima ullam unde, nostrum, molestias est enim, deleniti cumque aliquam laudantium temporibus! Facere nihil perferendis pariatur blanditiis. Soluta corrupti dolorem fugit?
About This Page Lorem ipsum dolor sit amet, consectetur adipisicing elit. Voluptate in incididunt tempora nemo dicta dolorem alias? Libero praesentium, unde odit itaque natus eum omnis id, molestias nesciunt non error nostrum?.

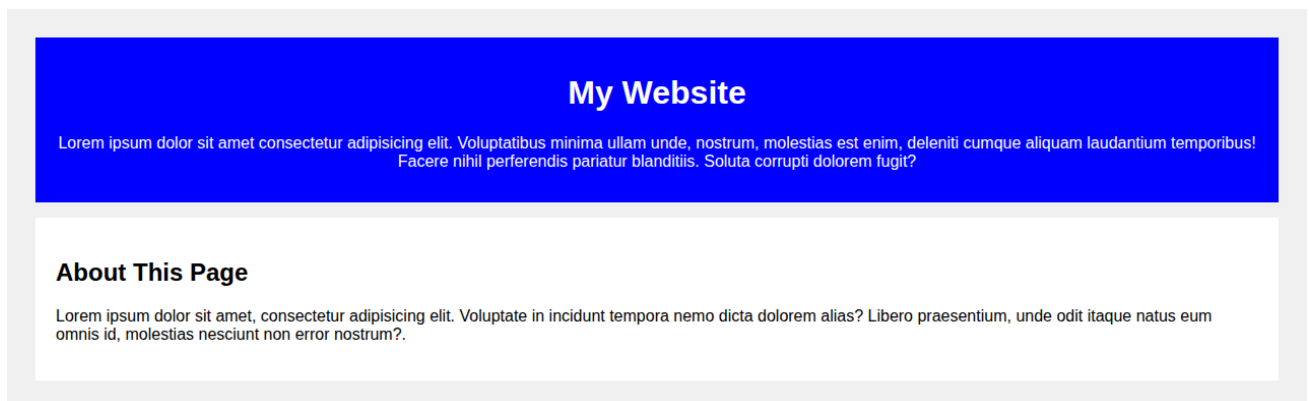
Test Case	TC-05
Test / Input	Check spacing between the header and content boxes
Expected Result	A 15px gap (margin-top) should appear between the header box and the content box.
Actual Result	A gap of 15px was measured between the two boxes.
Status	Pass

TC-05 Output:



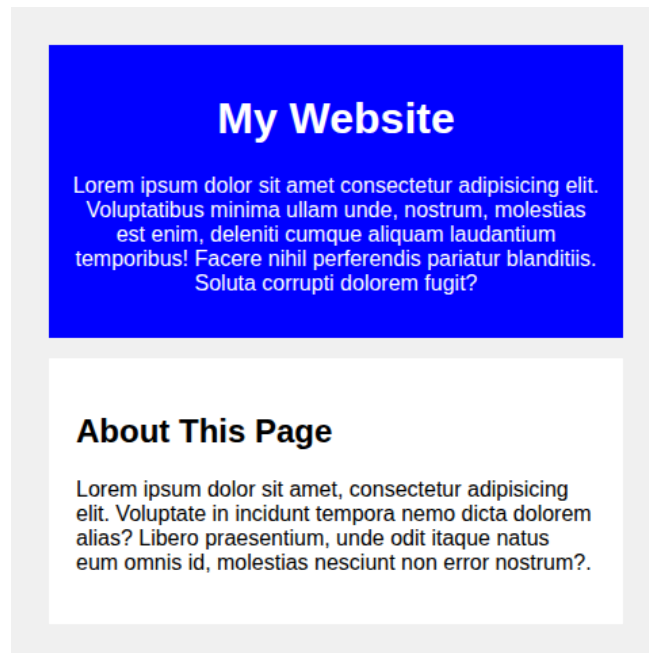
Test Case	TC-06
Test / Input	Check text alignment
Expected Result	Header text should be center aligned and content box text should be left aligned.
Actual Result	Header heading and paragraph were centered; content box heading and paragraph were left aligned.
Status	Pass

TC-06 Output:



Test Case	TC-07
Test / Input	Resize the browser window to a narrow width (480px)
Expected Result	The text should wrap and both containers should remain readable without breaking the layout.
Actual Result	Text wrapped onto more lines and both containers adjusted to the narrow width and remained readable.
Status	Pass

TC-07 Output:



Test Case	TC-08
Test / Input	Verify the applied CSS values using the browser computed styles
Expected Result	Computed values should match the CSS written in the code.
Actual Result	All computed values matched the CSS code, including the page title "Very Basic Page".
Status	Pass

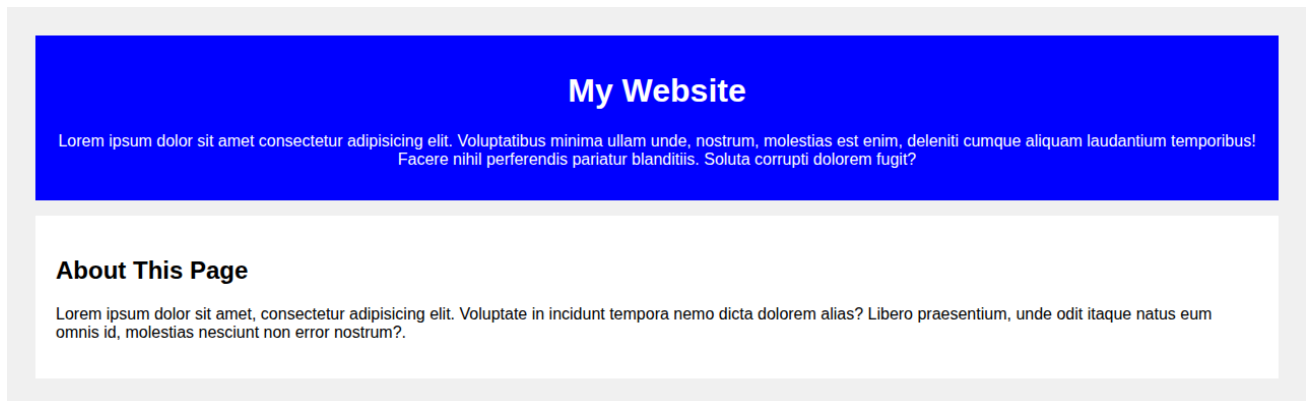
TC-08 Output:

Element	Property	Computed value (browser)
document	title	Very Basic Page
body	font-family	Arial, sans-serif
body	background-color	rgb(240, 240, 240)
body	padding / display	20px / inline-block
.header-box	background / color	rgb(0, 0, 255) / rgb(255, 255, 255)
.header-box	padding / text-align	15px / center
.content-box	background-color	rgb(255, 255, 255)
.content-box	padding / margin-top	20px / 15px

9. OUTPUT

Output: Complete Webpage

The following screenshot shows the complete execution of the HTML and internal CSS webpage in the web browser. It contains the blue header container with the heading "My Website" and the white content container with the heading "About This Page".



10. RESULT

Thus, a basic webpage was successfully designed and developed using HTML and internal CSS. The webpage contains a header container and a content container styled using class selectors, with background colors, text colors, padding, margin and text alignment applied. The webpage was executed and verified successfully in a web browser.